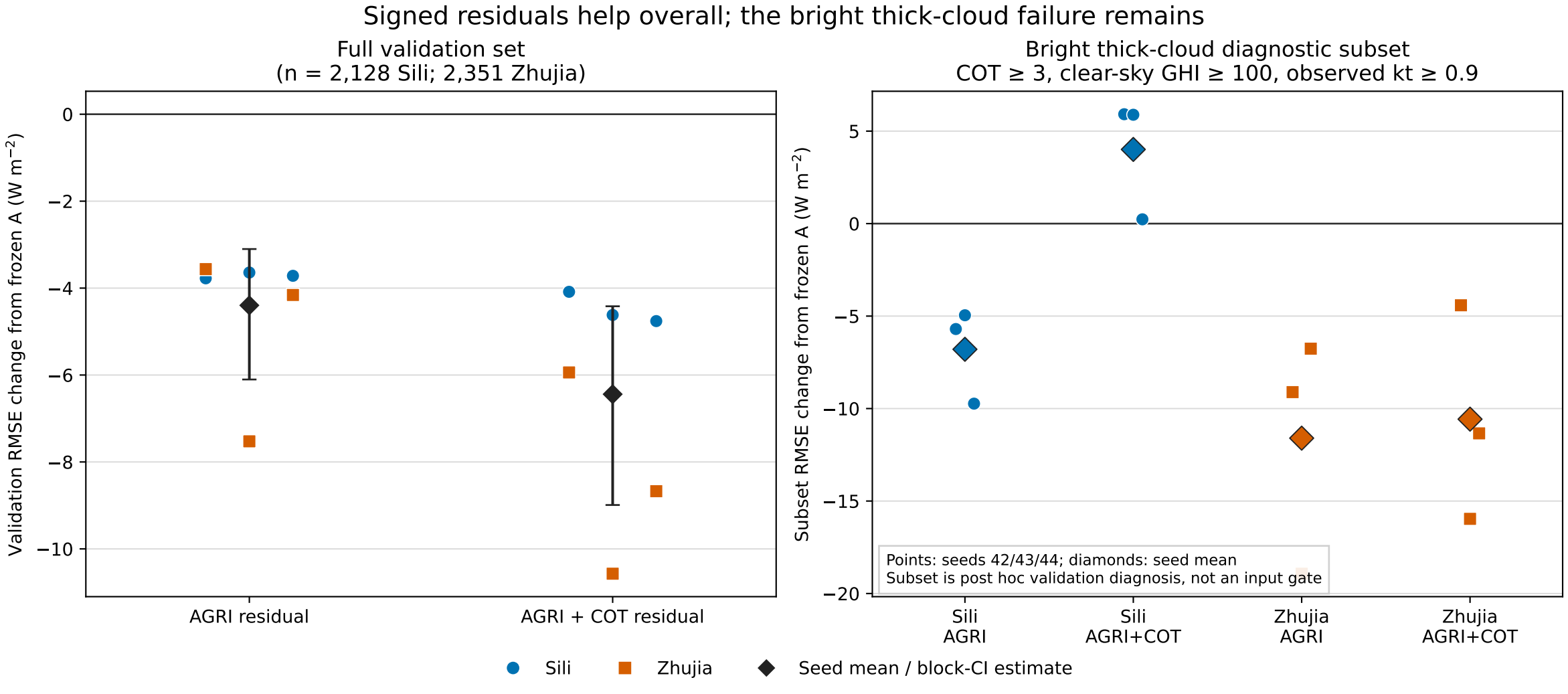
Signed residuals help overall; the bright thick-cloud failure remains
Full validation set
(n = 2,128 Sili; 2,351 Zhujia)
Validation RMSE change from frozen A (W m⁻²)
0
−2
−4
−6
−8
−10
AGRI residual
AGRI + COT residual
Bright thick-cloud diagnostic subset
COT ≥ 3, clear-sky GHI ≥ 100, observed kt ≥ 0.9
Subset RMSE change from frozen A (W m⁻²)
5
0
−5
−10
−15
−20
Points: seeds 42/43/44; diamonds: seed mean
Subset is post hoc validation diagnosis, not an input gate
Sili AGRI
Sili AGRI+COT
Zhujia AGRI
Zhujia AGRI+COT
Sili
Zhujia
Seed mean / block-CI estimate